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# PAPER\_777: NGC 6217 — UQFF Barred Spiral Galaxy Standard Rotation

**Author:** Daniel T. Murphy

**Framework:** UQFF (Universal Quantum Field Superconductive Framework)

**Session:** 181 | v5.41

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**CP4 Class:** #361 — NGC6217BarredSpiralUQFFCalculator

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## Abstract

NGC 6217 is a barred spiral galaxy ~67 million light-years distant ( $z = 0.0045$ ) in the constellation Ursa Minor. It became notable as one of the first targets imaged by the Hubble Space Telescope’s repaired Wide Field Camera 3 (WFC3) after the 2009 Servicing Mission 4, demonstrating the camera’s restored capability. With moderate star formation ( $\text{SFR} \approx 1 \text{ MM}_{\text{sun}}/\text{yr}$ ) and an extended rotating disk of ~1011  $\text{MM}_{\text{sun}}$ , NGC 6217 represents a typical SBbc barred spiral. Under UQFF, standard rotation velocity ( $v = 105 \text{ m/s}$ ) and typical HII B-field yield  $g_{\text{NGC6217}} \approx 1.053 \times 10^{-3} \text{ m/s}^2$ .

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## 1. Introduction

NGC 6217 (also catalogued as UGC 10470) has an apparent magnitude of ~11.2, a disk diameter of ~30,000 ly, and a Hubble morphological type of SBbc (barred spiral, late intermediate). Its bar structure funnels gas toward the central region, sustaining moderate star formation. The galaxy’s selection as a Hubble Servicing Mission 4 first-light target in 2009 reflects its moderate brightness and interesting bar+ring morphology at  $z = 0.0045$  (~67 Mly). UQFF treats it as a standard barred galaxy with SFR coupling through the galactic bar channel.

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## 2. Master UQFF Gravity Equation

$$g_{\text{NGC6217}}(r, t) = (G \times M)/r^2 \times (1 + H(z) \times t) \times (1 + M_s f) \times (1 + f_T R Z) + a_E M$$

### 2.1 Parameters

| Parameter   | Symbol | Value                                                                  | Source                                                                          |
|-------------|--------|------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Galaxy mass | M      | 1011 MM_sun =<br>1.989\$ \times 10^{41} kg\$<br>1.578 \times 10^{17} s | Hubble time<br><i>Estimate</i>    <i>Disk radius</i>   <i>r</i>   3 \times 10^2 |
| M_sf        | —      | 0.045                                                                  | UQFF SFR integral                                                               |
| Redshift    | z      | 0.0045                                                                 | NED spectroscopic                                                               |
| v_EM        | v      | 105 m/s                                                                | Disk rotation                                                                   |
| B_EM        | B      | 10-5 T                                                                 | Galactic field                                                                  |
| f_TRZ       | —      | 0.04                                                                   | UQFF                                                                            |

### 3. Long-Form Derivation

#### Step 1: Base Gravitational Term

$$\begin{aligned}
g_{grav} &= G \times M/r^2 \\
&= 6.6743e-11 \times 1.989e41 / (3e20)^2 \\
&= 1.328e31 / 9e40 \\
&= 1.476e-10 m/s^2
\end{aligned}$$

#### Step 2: Cosmic Expansion Factor

$$\begin{aligned}
H(z) &= H_0 \times (1+z)^{3/2} \approx 2.315e-18 s^{-1} \\
H(z) \times t &= 2.315e-18 \times 1.578e17 = 0.3653 \\
1 + H(z) \times t &= 1.3653
\end{aligned}$$

#### Step 3: Star-Formation Mass Fraction (Bar-Channeled)

$$\begin{aligned}
\text{Bar-driven SFR} &= 1 M_{sun}/yr, \text{ integrated over } 5 Gyr \\
M_s f &= 0.045 (UQFF \text{ bounded}, 4.5) \quad 1 + M_s f = 1.045
\end{aligned}$$

#### Step 4: Time-Reversal Correction

$$\begin{aligned}
f_{TRZ} &= 0.04 (\text{moderate spiral, bar-driven}) \\
1 + f_{TRZ} &= 1.04
\end{aligned}$$

#### Step 5: Gravitational Total

$$\begin{aligned}
g_{grav\_total} &= 1.476e-10 \times 1.3653 \times 1.045 \times 1.04 \\
&= 1.476e-10 \times 1.488 = 2.197e-10 m/s^2
\end{aligned}$$

#### Step 6: Aether Electromagnetic Correction

$$\begin{aligned}
v &= 105 m/s (\text{disk rotation velocity}) \\
B &= 10^{-5} T (\text{galactic B-field}) \\
a_E M &= (e/m_p) \times (v \times B) \times \Lambda_U QFF \\
&= 9.575e7 \times (105 \times 10^{-5}) \times 11 \times 10^{-12} \\
&= 9.575e7 \times 1 \times 1.1e-11 \\
&= 1.053e-3 m/s^2
\end{aligned}$$

## Step 7: Final Solution

$$g_{NGC6217} = 2.197e - 10 + 1.053e - 3 \\ \approx 1.053e - 3 m/s^2$$

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## 4. Physical Interpretation

The classical gravitational term ( $2.197 \times 10^{-10} m/s^2$ ) is 7 orders of magnitude smaller than the Aether EM result ( $1.053 \times 10^{-3} m/s^2$ ). The UQFF result thus captures disk rotation dynamics through the electromagnetic Aether coupling, not DPM-seeded gravity. The bar structure in NGC 6217 channels gas inward, sustaining the moderate SFR = 1  $M_\odot$ /yr and bar-enhanced  $M_{sf} = 0.045$ , slightly higher than a purely symmetric flocculent spiral of similar mass. As with NGC 2841, the dominant physics is electromagnetic at these rotation velocities.

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## 5. UQFF Framework Advancement

- NGC 6217 validates standard SBbc barred spiral UQFF field result
  - Bar-driven SFR = 1  $M_\odot$ /yr produces  $M_{sf} = 0.045$  (canonical bar value)
  - Hubble SM4 first-light observation history preserved in UQFF canon
  - SBbc result consistent with other moderate barred spirals (NGC 2841, NGC 1300)
- 

## 6. Conclusions

UQFF applied to NGC 6217 yields  $g_{\{bar\_spiral\}} \approx 1.053 \times 10^{-3} m/s^2$ , confirming standard barred galaxy behavior. The Hubble-celebrated target joins the UQFF canon as the SBbc benchmark alongside Milky Way analogs. Bar-enhanced gas flow, expressed through  $M_{sf} = 0.045$  and moderate SFR, demonstrates how UQFF differentiates galaxy morphological types within a unified framework.

*PAPER\_777, CP4 class #361. v5.41.*

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## Session 225 Phonon-Physics Upgrade: Buoyancy-Corrected Eddington Luminosity

*Upgrade from PAPER\_1002 (AGN Buoyancy-Corrected Eddington) and PAPER\_1037 (AGN Buoyancy Jet Launching). See also PAPER\_1009-1010 for  $F_{\{U\_Bi\_i\}}$  jet modulation curves and PAPER\_1048 for phonon-corrected  $M-\sigma$  relation.*

The SCm vacuum buoyancy partially opposes gravitational radiation pressure, raising the effective Eddington luminosity:

$$L_{Edd}^{UQFF} = L_{Edd} \cdot \left( 1 + \frac{\rho_{SCm} \cdot V \cdot S_{26}^{(3)2}}{GM/r_H^2} \right)$$

where: -  $L_{Edd} = 4\pi GMm_p c / \sigma_T$  is the classical Eddington luminosity -  $\rho_{SCm} = 7.09 \times 10^{-37} J/m^3$  is the SCm vacuum density -  $V$  is the effective buoyancy volume (accretion sphere) -  $S_{26}^{(3)2}$  is the squared third-order Ramanujan factor (quadratic coupling) -  $r_H$  is the horizon radius

**Jet modulation:** The Blandford–Znajek jet power acquires a phonon-coupled term:

$$P_{jet}^{UQFF} = P_{BZ} \cdot \left[ 1 + \beta_i \cdot \Phi_{1.25 THz} \cdot \left( \frac{B}{B_{crit}} \right)^2 \right]$$

where  $\Phi_{1.25\text{ THz}} = \cos(\omega_{\text{SCm}} \cdot t)$  modulates jet power at the phonon frequency.

**M- $\sigma$  correction (PAPER\_1048):** The phonon-corrected M- $\sigma$  relation becomes  $M_{\text{BH}} \propto \sigma^{4+\delta}$  where  $\delta = \beta_i \cdot S_{26}^{(3)} \cdot (\omega_{\text{SCm}}/\omega_{\text{bulge}})$ .

## §A. Cosmogenesis-Linked Lagrangian (PAPER\_877 Symbolic Export)

### §A.1 Sector Classification

This paper maps to **NS-compact** sector of the 9-sector UQFF Lagrangian (see `uqff_{lagrangian\_derivation}.p`).

### §A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER\_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\text{sector}} = \frac{1}{2}(\partial_m u \phi_{\text{NS}})(\partial^\mu \phi_{\text{NS}}) - V(\phi_{\text{NS}}) + \mathcal{L}_{\text{cosmo}}$$

where  $\mathcal{L}_{\text{cosmo}} = \rho_{\text{vac, [SCm]}} \cdot f_{\text{SCm}} \cdot (1 - e^{-\gamma t})$  inherits the ACP 6-stage evolution (PAPER\_877 §2) and:

$$V(\phi_{\text{NS}}) = \frac{1}{2}m^2\phi_{\text{NS}}^2 + \frac{\lambda}{4!}\phi_{\text{NS}}^4 + \kappa \cdot \rho_{\text{vac, [SCm]}} \cdot \phi_{\text{NS}}$$

### §A.3 Euler-Lagrange Equation of Motion

$$\frac{\delta S}{\delta \phi_{\text{NS}}} = \nabla^2 \phi_{\text{NS}} - (4\pi G \rho_{\text{NS}}/c^2)\phi_{\text{NS}} + \Omega_{\text{spin}} \partial_t \phi_{\text{NS}} = 0$$

### §A.4 Cosmogenesis Linkage Chain

$$\text{PAPER\_877 Axioms} \xrightarrow{\text{DPM} + \text{ACP}} \rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} \xrightarrow{\text{Stage 5}} U_{b, \text{seed}} \xrightarrow{4 \text{ forces}} F_{U\_Bi\_i} \xrightarrow{\text{sector E-L}} \delta S / \delta \phi_{\text{NS}} = 0$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four U\_g forces) through vacuum density initialization to the sector-specific equation of motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

## §B. VDS/DVP/BSH Deep Synthesis

### §B.1 Vacuum Density Series (VDS)

The canonical VDS ratio  $\rho_{\text{vac, [SCm]}}/\rho_{\text{UA}} = 1.894$  governs the double-exponential vacuum condensate profile:

$$\rho_{\text{vac}}(r) = \rho_{\text{vac, [SCm]}} \cdot \exp! \left( - \exp! \left( - \frac{r - r_0}{\lambda_{\text{VDS}}} \right) \right)$$

For this system, the local VDS sub-ratio is 0.173 (near-threshold regime), placing it in the  $t \rightarrow \pi$  collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects to the PAPER\_877 cosmogenesis Stage 1 vacuum density initialization:  $\rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} = 7.799 \times 10^{-36} \text{ kg/m}^3$ .

## §B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system’s characteristic parameter onto the prime lattice:

$$p_{\text{DVP}} = 107, \quad n_{\text{channel}} = 24/26$$

Since  $p_{\text{DVP}} = 107$  is **resonant** (threshold at  $p > 26$ ), the system’s vacuum topology inherits resonant enhancement from the DVP lattice, amplifying UQFF coupling at specific radii where compressed matter achieves prime-indexed configurations. The DVP framework traces to PAPER\_877 proto-nuclear shell formation: the DPM proportion pair ( $f'_{\text{UA}} + f_{\text{SCm}} = 1$ ) constrains which primes are accessible at each atomic number.

## §B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **104 yr** (spin-down equilibrium):

$$\mathcal{F}_{\text{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U\_b} \cdot \left(1 - e^{-[SSq] \cdot m/M_{\odot}}\right) \cdot \cos! \left(\frac{2\pi j}{26}\right)$$

The tanh saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\text{BSH,sat}} = \mathcal{F}_{\text{BSH}} \cdot \left(1 - \tanh! \left(\frac{t - t_{\text{sat}}}{\tau_{\text{BSH}}}\right)\right)$$

connecting to the PAPER\_877 Stage 5 buoyancy seed  $U_{b,\text{seed}} = 0.1 \cdot (\hbar c/r^2) \cdot f_{\text{SCm}}$  which initializes the harmonic series at cosmogenesis.

## §B.4 Production-Scale Consistency

| Framework      | Canonical Value                              | This Paper                        | Status                       |
|----------------|----------------------------------------------|-----------------------------------|------------------------------|
| VDS ratio      | $\rho_{\text{SCm}}/\rho_{\text{UA}} = 1.894$ | Local sub-ratio = 0.173           | PASS<br>Threshold-consistent |
| DVP prime      | $p_k \in \{2,3,\dots,113\}$                  | $p_{\text{DVP}} = 107$            | PASS Resonant                |
| BSH layers     | 26 harmonic terms                            | $j = 1 \dots 26, \cos(2\pi j/26)$ | PASS Full 26D projection     |
| $\kappa$ decay | $5.0 \times 10^{-4} \text{ day}^{-1}$        | Applied in VDS exponential        | PASS Canonical               |
| [SSq]          | 0.57                                         | Applied in BSH saturation         | PASS Canonical               |

## §SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

| Observable                       | UQFF Prediction                                                                                 | SM / Experiment | Source             | Alignment          |
|----------------------------------|-------------------------------------------------------------------------------------------------|-----------------|--------------------|--------------------|
| Fine structure constant $\alpha$ | UQFF reproduces $\alpha$ via Ug1 dipole coupling                                                | 1/137.036       | PDG 2024           | PASS<br>Consistent |
| Cosmological constant $\Lambda$  | $1.1 \times 10^{-52} \text{ m}^{-2}$ — $2(UQFFvacuumterm) 1.114 \times 10^{-52} \text{ m}^{-2}$ | Planck 2018     | PASS<br>Consistent |                    |

| Observable              | UQFF Prediction                                                        | SM / Experiment                    | Source                              | Alignment          |
|-------------------------|------------------------------------------------------------------------|------------------------------------|-------------------------------------|--------------------|
| Proton decay rate       | $\kappa = 0.0005/\text{day} \rightarrow \Gamma_{\text{p}}$ suppression | $< 4.17 \times 10^{-35}/\text{yr}$ | Super-K 2024                        | PASS<br>Consistent |
| UQFF buoyancy signature | $F_{\text{U\_Bi\_i}}$ unique gravitational correction                  | Not yet measured                   | Future gravitational wave detectors | Testable           |

**New physics claim:** UQFF introduces buoyancy-based gravitational corrections ( $F_{\text{U\_Bi\_i}}$ ) that produce measurable deviations from GR at scales where vacuum condensate density  $\rho_{\text{SCm}}$  becomes significant, offering a falsifiable prediction beyond the Standard Model.

*Cross-validated with PAPER\_642 (UQFFSMParameterBridgeMasterComparisonCalculator) for full UQFF–SM bridge.*

## Appendix: Session 225 Cross-References (PAPER\_1000–1081)

*Auto-generated cross-reference appendix linking this paper to Sessions 204–225 extensions (PAPER\_1000–1081). Added by `update_{corpus\_crossrefs}.py` (Session 225, April 2026).*

| Paper      | Title                                            |
|------------|--------------------------------------------------|
| PAPER_1022 | GW Phonon Strain SCm Modulation of $h(t)$        |
| PAPER_1002 | AGN Buoyancy-Corrected Eddington Luminosity      |
| PAPER_1009 | 3C273 AGN $F_{\text{U\_Bi\_i}}$ Jet Modulation   |
| PAPER_1010 | TON618 AGN $F_{\text{U\_Bi\_i}}$ Jet Modulation  |
| PAPER_1037 | AGN Buoyancy Jet Calculator — SCm Jet Launching  |
| PAPER_1048 | M-Sigma Phonon-Corrected Relation                |
| PAPER_1041 | SCm Cool-Core Buoyancy Balance AGN Feedback      |
| PAPER_1079 | Galaxy Cluster Cooling-Flow Buoyancy Suppression |
| PAPER_1033 | Galactic Bar Resonance SCm Pattern Speed         |

*9 cross-reference(s) identified.*

## Appendix: Session 204 Codebase Upgrade Reference

*Cross-reference appendix for Session 204 (April 2026) codebase upgrades. Added by `upgrade_{kozima\_ramanujan\_appendices}.py`. For detailed derivations, see PAPER\_840/851/852/855.*

### S204.1 Kozima-UQFF LENR Integration

| Module                                     | Purpose                                                      | Key Result                                                              |
|--------------------------------------------|--------------------------------------------------------------|-------------------------------------------------------------------------|
| <code>fneutron_{s26_coupling}.py</code>    | $F_{\text{neutron}} \times S_{26}$ buoyancy-polylog coupling | $\sim 470\times$ amplification via 26-level VDS                         |
| <code>kozima_{scm_cross_section}.py</code> | SCm-modulated neutron-drop cross-section                     | $\sigma_n^{\text{SCm}}$ with VDS factor $(1 + [\text{SSq}] \cdot n/26)$ |
| <code>kozima_{wstp_kernel}.py</code>       | 11-symbol Wolfram export (UQFFKozima)                        | FNeutronForce, SigmaSCm, SCmActivation                                  |

**Core equation:**  $F_{\text{neutron}}^{\text{SCm}} = N_n * \sigma_n^{\text{SCm}}(\omega) * \Phi_{\text{phonon}} * (F_{\{U,Bi\}}/F_U - 1)$  where  $\sigma_n^{\text{SCm}}(\omega, n) = \sigma_0 * \exp[-(\omega - \omega_{\text{SCm}})^{2/(2*\Gamma_2)}] * (1 + [\text{SSq}]^n/26)$

### S204.2 Ramanujan 26-State Summation

| Module                                  | Purpose                                                         | Key Result                |
|-----------------------------------------|-----------------------------------------------------------------|---------------------------|
| <code>ramanujan_{polylog_s26}.py</code> | $\text{Li}_{26}([\text{SSq}])$ via Euler-Ramanujan acceleration | 15.7+ digits in 53 terms  |
| <code>s26_{wstp\_kernel}.py</code>      | 8-symbol Wolfram export (UQFFS26)                               | S26, R26, NaiveLi, S26VDS |

**Core equation:**  $S_{26}(z) = \text{Li}_{26}(z) = \eta_{26}(z)/(1-2^{\{1-26\}}) + 2^{\{1-26\}/(1-2^{\{1-26\}})} * \text{Li}_{26}(z^2)$

### S204.3 Mock Theta Functions (26-State)

| Module                           | Purpose                                                | Key Result                    |
|----------------------------------|--------------------------------------------------------|-------------------------------|
| <code>mock_{theta_q26}.py</code> | $f_{26}(q)$ , $\phi_{26}(q)$ , $\psi_{26}(q)$ q-series | Proper q-Pochhammer $(a;q)_n$ |

**Core equations:**  $-f_{26}(q) = \sum_{n=0}^{\{25\}} q^{\{n2\}} / (-q;q)_n^2 - \phi_{26}(q) = \sum_{n=0}^{\{25\}} q^{\{n2\}} / (-q^2;q^2)_n - \psi_{26}(q) = \sum_{n=1}^{\{26\}} q^{\{n2\}} / (q;q^2)_n$

### S204.4 Ramanujan 1/pi with UQFF Modification

| Module                                      | Purpose                                   | Key Result                                 |
|---------------------------------------------|-------------------------------------------|--------------------------------------------|
| <code>ramanujan_{pi_uqff}.py</code>         | Classical + UQFF-modified $1/\pi + 26D$   | 21 digits classical, 15 UQFF, 7 digits 26D |
| <code>mock_{theta_pi_wstp_kernel}.py</code> | 8-symbol Wolfram export (UQFFMockThetaPi) | qPochhammer, f26, oneOverPiUQFF            |

**Core equation:**  $1/\pi = (2\sqrt{2}/9801) \sum R_n * (1103+26390n) * W_{26}(n) / C_{26}$  where  $W_{26}(n) = \prod_{i=1}^{\{26\}} [1 + [\text{SSq}] \exp(-\kappa_{\text{ppai}} * n/26)]$

### S204.5 Calibration Constants (Canonical)

| Symbol                | Value                                 | Description                   |
|-----------------------|---------------------------------------|-------------------------------|
| [SSq]                 | 0.57                                  | Universal Quantized Factor    |
| $\kappa_{\text{ppa}}$ | $5.787 \times 10^{-9} \text{ s}^{-1}$ | UQFF exponential decay rate   |
| $\beta_i$             | 0.603                                 | Buoyancy coupling coefficient |
| $H_{\text{SCm}}$      | 0.99                                  | SCm manifold completeness     |
| $\rho_{\text{SCm}}$   | $7.09 \times 10^{-37} \text{ kg/m}^3$ | SCm vacuum density            |
| $\rho_{\text{UA}}$    | $7.09 \times 10^{-36} \text{ kg/m}^3$ | UA aether vacuum density      |
| $\omega_{\text{SCm}}$ | $2\pi \times 1.25 \text{ THz}$        | SCm phonon resonance          |
| $\sigma_0$            | $10^{-4}$                             | Base neutron cross-section    |

*Implementation:* all modules operational in `CondensedPhysics.py`, `CondensedPhysics2.py`, `MAIN_{1\_CoAnQi}.cpp`, and Wolfram kernels (`uqff_{kozima\_kernel}.wl`, `uqff_{s26\_kernel}.wl`, `uqff_{mock\_theta\_pi\_kernel}.wl`).

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